Multi-file Program

Ch 13

Object Oriented Programming

CSE-4301

Presented by

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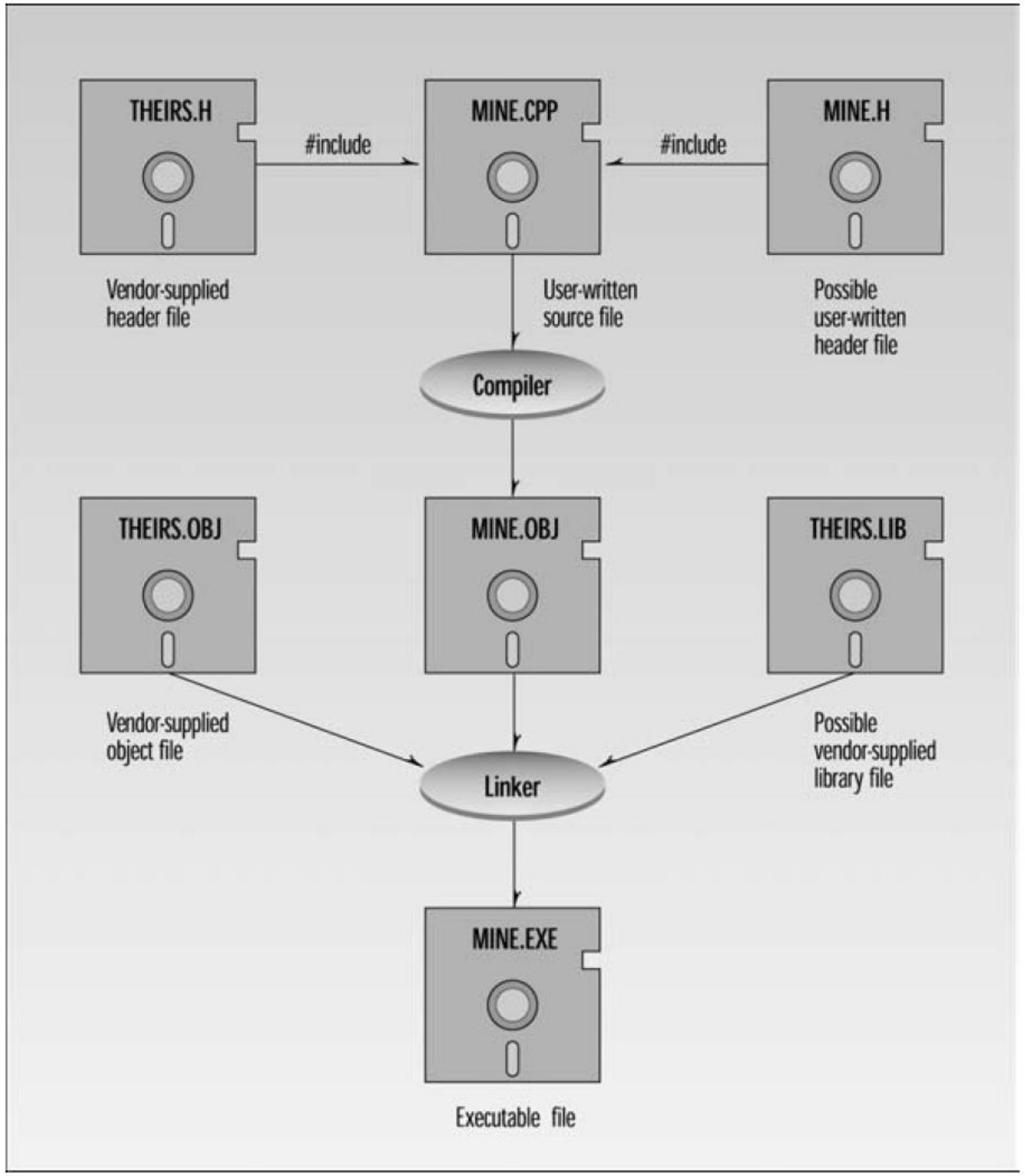
Reasons for Multifile Programs

- Class Libraries
  - Vendors provide well furnished library of functions
  - Cpp contains library of classes which is better in modeling
  - It is composed of two components
    - Interface
    - Implementation

Multifile System



Interfile Communication

- Cpp files are separately compiled
- Its important to understand how these files communicated with each other
- How header files play its role

Distinction between declaration and definition

- Declaration is kind of announcement that this program element is available but how that program element will function is not present in declaration.

- Definition means that how a program element will function.

- Declaration and Definition of
  - Variable
  - Function
  - Class

Interfile Variables

As you might expect, a global variable can be defined in only one place in a program.

```
//file A
int globalVar;  //definition in file A

//file B
int globalVar;  //illegal: same definition in file B
```

```
//file A
int globalVar;   //defined

//file B
globalVar = 3;   //illegal, globalVar is unknown here
```

Interfile Variables

```
//file A
int globalVar;           //definition

//file B
extern int globalVar;    //declaration
globalVar = 3;           //now this is OK
```

```
extern int globalVar = 27;  //not what you might think
```

```
//file A
static int globalVar;   //definition; visible only in file A

//file B
static int globalVar;   //definition; visible only in file B
```

Interfile Function

```
//file A
int add(int a, int  b)    //function definition
   { return a+b; }        //(includes function body)

//file B
int add(int, int);        //function declaration (no body)
. . .
int answer = add(3, 2);   //call to function
```

```
//file A
int add(int, int);  //declaration
int add(int, int);  //another declaration is OK
```

Like variables, functions can be made invisible to other files by declaring them `static`.

```
//file A
static int add(int a, int  b)   //function definition
   { return a+b; }

//file B
static int add(int a, int  b)   //different function
   { return a+b; }
```

This code creates two distinct functions. Neither is visible in the other file.

Interfile Classes

A class declaration is simply a statement that a certain name applies to a class. It conveys no information to the compiler about the members of the class.

```
class someClass;                //class declaration
```

A class definition contains declarations or definitions for all its members:

```
class someClass                  //class definition
   {
   private:
      int memVar;                //member data definition
   public:
      int memFunc(int, int);     //member function declaration
   };
```

Why class definition is required

Why does a class need to be defined in every file where it's used? The compiler needs to know the data type of everything it's compiling. A declaration is all it needs for simple variables because the declaration specifies a type already known to the compiler.

```
//declaration
extern int someVar;            //if it sees this, the compiler
someVar = 3;                   //can generate this
```

```
//declaration
int someFunc(int, int);        //if it sees this, the compiler
var1 = someFunc(var2, var3);   //can generate this
```

However, for a class, the entire definition is necessary to specify the types of its member data and functions.

```
//definition
class someClass                //if it sees this, the compiler
   {
   private:
      int memVar;
   public:
      int memFunc(int, int);
   };
someClass someObj;             //can generate this
v1 = someObj.memFunc(v2, v3);  //and this
```

Header files

```
//fileH.h
extern int gloVar;                //variable declaration
int gloFunc(int);                 //function declaration
```

```
//fileA.cpp
int gloVar;                       //variable definition
int GloFunc(int n)                //function definition
   { return n; }

//fileB.cpp
#include "fileH.h"
. . .
gloVar = 5;                       //work with variable
int gloVarB = gloFunc(gloVar);    //work with function
```

```
//fileH.h
class someClass                          //class definition
   {
   private:
      int memVar;
   public:
      int memFunc(int, int);
   };

//fileA.cpp
#include "fileH.h"
int main()
   {
   someClass obj1;                       //create an object
   int var1 = obj1.memFunc(2, 3);        //work with object
   }

//fileB.cpp
#include "fileH.h"
int func()
   {
   someClass obj2;                       //create an object
   int var2 = obj2.memFunc(4, 5);        //work with object
   }
```

Multiple Include Hazard

```
//file app.cpp
#include "headone.h"
#include "headone.h"
```

```
//file headtwo.h
int globalVar;

//file headone.h
#include "headtwo.h"

//file app.cpp
#include "headone.h"
#include "headtwo.h"
```

```
//file app.cpp
. . .
int globalVar;  //from head2.h via headone.h
. . .
int globalVar;  //from head2.h directly
```

This will cause the compiler to complain that `globalVar` is defined twice.

Prevention of Multiple Includes

```
#if !defined( HEADCOM )        //if HEADCOM not defined,
#define HEADCOM                //define it

int globalVar;                 //define this variable


int func(int a, int b)         //define this function
   { return a+b; }

#endif                         //end condition
```

Namespace

- To avoid writing unique identifier name
- Namespace can be declared multiple times
- Namespace is normally used in header file

```
namespace geo
   {
   const double PI = 3.14159;
   double circumf(double radius)
      { return 2 * PI * radius; }
   }  //end namespace geo
```

```
namespace geo
   {
   const double PI = 3.14159;
   double circumf(double radius)
      { return 2 * PI * radius; }
   }  //end namespace geo

double c = circumf(10);  //won't work here
```

```
double c = geo::circumf(10);  //OK
```

Or you can use the using directive:

```
using namespace geo;
double c = circumf(10);  //OK
```

typedef

- Using typedef you can rename any types
  - Type of variable
  - Class name

```
typedef unsigned long unlong;

unlong var1, var2;
```

```
typedef int FLAG;        //int variables used to hold flag values
typedef int KILOGRAMS;   //int variables used to hold values in kilograms
```

If you don't like the way pointers are specified in C++, you can change it:

```
int *p1, *p2, *p3;       //normal declaration
typedef int* ptrInt;     //new name for pointer to int
ptrInt p1, p2, p3;       //simplified declaration
```

This avoids all those pesky asterisks.

```
class GeorgeSmith_Display_Utility          //class definition
   {
   //members
   };

typedef GeorgeSmith_Display_Utility GSdu;   //rename the class

GSdu anObj;                                 //create object using new name
```

Thank You

Any questions?

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